

Weekly AI Brief

23–29 September 2026

Papers, model releases, benchmarks and policy from the past seven days

Prepared by Ali Habibullah, KAUST

The week in five lines

- **Claude Sonnet 5.5 ships at \$2/\$10 per million tokens.** Artificial Analysis independently places it 3rd of 216 models, two points behind Opus 5.5. [S1] [S2]
- **OpenAI halved its mid-tier prices.** GPT-6 Sol and Luna launched on 22 September at 50% below GPT-5.6 pricing, about 90 minutes after Anthropic's Opus 5.5. [S3] [S5] [S19]
- **Xiaomi posted MIT-licensed 1.02T-parameter MiMo-V2.6 checkpoints.** The 27 September update targets agents that repeat the same tool call. [S7]
- **Claude agents flagged a new CRISPR-like enzyme family in phage DNA.** Anthropic's own preprint says its function is still untested. [S9] [S10]
- **The UN Security Council heard from Altman, Amodei and Bengio.** The US told the Council that dialogue "cannot be allowed to drift towards global governance". [S15]

1. Model and product releases

Claude Sonnet 5.5: near-Opus scores at half the Opus price

Anthropic • API and all major clouds • 28 September • [link](#) • [S1]

Sonnet 5.5 (`claude-sonnet-5-5`) costs \$2 input and \$10 output per million tokens, with cache reads at \$0.20. It is available on AWS, Google Cloud and Microsoft Azure. Anthropic says it is 30%+ faster than Sonnet 5 and costs up to 30% less per task. Its largest self-reported jump is on Terminal-Bench 4.0, from 10.3% to 70.6%, which is above Opus 5.5's 66.4% on the same page. Artificial Analysis independently scores it 56 on Intelligence Index v4.3.2 (#3 of 216). It also lists a 1M-token context window and flags the model as "very verbose": 410M output tokens across the evaluation, against an 88M median. [S2]

Why it matters. For most agentic coding workloads, the default question is now whether Opus is worth twice the price, not whether Sonnet is good enough.

GPT-6 Sol and Luna: OpenAI halves prices below the GPT-6 flagship

OpenAI • API, ChatGPT Work, Codex • 22 September (one day before this window) • [link](#) • [S3]

Carried here because they launched on the last day of issue #05's window, after it was compiled. Sol drops from \$4/\$20 to \$2/\$10 and Luna from \$0.20/\$1.20 to \$0.10/\$0.50 per million tokens. OpenAI says both models were trained "with similar methods as GPT-6 Astra". It self-reports DeepSWE v1.1 at 68.8% for Sol (max effort), 1.1 points below Claude Fable 5 at "approximately 80% lower cost per task". On its internal factuality evaluation, Sol "makes about half as many

mistakes” as its predecessor. OpenAI notes that this evaluation is built from conversations where users flagged errors, so it is not representative of typical use. Cached input reads are discounted 90%. The API IDs are `gpt-6-sol` and `gpt-6-luna`. Artificial Analysis scores Sol (max) at 48 (#20 of 216) with an 872K context window. [S4]

Why it matters. Bulk extraction and summarisation jobs that were routed to small models on cost grounds should be re-priced against Luna at \$0.10/\$0.50.

Claude Opus 5.5 tops the independent index

Anthropic • API and all major clouds • 22 September (one day before this window) • [link](#) • [S5]

Also a 22 September launch that issue #05 missed. Opus 5.5 costs \$4/\$20 per million tokens (\$8/\$40 in fast mode), which Anthropic says is “40% less to run than Opus 5” on typical workloads. Its self-reported table puts it ahead of GPT-6 Astra on Terminal-Bench 4.0 (66.4% vs 57.9%) and behind Astra on AutomationBench (40.0% vs 41.4%) and Terminal-Bench-Science (58.7% vs 64.6%). Artificial Analysis independently ranks it #1 of 216 on Intelligence Index v4.3.2, at 58. [S6]

Why it matters. The model at the top of the independent index is also, by Anthropic’s account, 40% cheaper to run than Opus 5, so teams that stayed on Fable 5.1 for quality should re-run their evals on Opus 5.5.

MiMo-V2.6 open weights, 1.02T parameters, MIT licence

Xiaomi • Hugging Face and ModelScope • MOPD checkpoints posted 27 September • [link](#) • [S7]

Pro is a sparse MoE with 1.02T total and 42B active parameters. Flash has 309B total and 15B active. Both have a 1M-token context and native text, image, video and audio input. The RL checkpoints appeared on 21–22 September. The in-window 27 September “MOPD” checkpoints add multi-teacher on-policy distillation and target tool-call repetition, a failure mode where an agent “keeps issuing the same or highly similar tool calls, appearing busy while making no progress”. Xiaomi’s self-reported table for the RL checkpoint puts Pro at 71.9 on DeepSWE v1.1, against 74.0 for Claude Opus 5. [S8]

Why it matters. A trillion-parameter checkpoint under MIT removes the licence question for fine-tuning and commercial use, and the repetition failure mode it names is worth checking for in any agent harness.

2. Research worth reading

Claude agents flag a CRISPR-like reverse-transcriptase family

Yoon, Athukoralage, Ameisen et al., Anthropic • preprint, 23 September • [link](#) • [S9]

Claude Code agents surveyed reverse-transcriptase loci across 1.9 billion protein clusters. They surfaced “array-associated RTs” (ART): an RT, a dedicated partner gene and a long array of roughly 200-nucleotide repeat units, found in bacteriophage DNA. The preprint reports 95 distinct ART RT clusters, 28 of them with detectable upstream arrays. [S10] Anthropic’s post puts the campaign at about 950 agents, 21 hours and 210 million tokens. The preprint is explicit that “ART’s biochemical activity and biological function remain untested”. What the agents did was anomaly detection, not a demonstrated mechanism.

Why it matters. It is a concrete, checkable case of an agent campaign surfacing an uncharacterised enzyme family; whether it matters biologically depends on functional assays the preprint says have not been done.

Just-in-Time Memory: curate agent memory at read time, not write time

Zhou, Li, Liu, Yavuz, Joty • arXiv:2609.27334 • 23 September • [link](#) • [S11]

Most agent memory systems compress each finished trajectory into a fixed reflection or skill before

anyone knows what future task will need it. JitMem keeps raw trajectories. A trained curator writes a task-specific payload only when a new task arrives, which also lets it learn from immediate task success. Over the strongest baseline it gains 16.2 (ALFWorld), 16.3 (WebShop) and 3.9 (τ^2 -bench) absolute success-rate points. The arXiv page links no code.

Why it matters. If you are building agent memory, storing raw traces and summarising on retrieval is cheap to try and beat learned write-time summaries here.

A measurable definition of “interesting” theorems

Patel, Rammal, Hayat, Munos, Kempe • arXiv:2609.28603 • 23 September • [link](#) • [S12]

The authors define a theorem’s interestingness as the ratio of proof length to statement length. They train a 27B model to predict proof difficulty and use it to select LLM-generated theorems for a growing machine-verified library. Selection cuts overlap with existing Mathlib from 91.9% to 30.6%. No code link is given on the arXiv page.

Why it matters. Automated maths has had a verifier but no filter; a cheap, checkable proxy for novelty is the missing piece for self-expanding formal libraries.

Transformers can hold two next-token distributions at once

Tikhonov, Korznikov, . . . , Oseledets, Tutubalina • arXiv:2609.29845 • 24 September • [link](#) • [S13]

When inputs from two text streams are linearly combined, the model outputs a superposition of the two next-token distributions. The authors find this property is architectural rather than learned, and that training weakens it. Fine-tuning restores it, and a new decoding method produces two coherent continuations from a single forward pass. No code is linked.

Why it matters. If this reproduces at scale, it is a new route to batched or speculative generation that needs no extra model.

PISA: block-sparse attention in $O(N \log N)$

Tang, Qin, Pan, Li, Liu • arXiv:2609.31093 • 25 September • [link](#) • [S14]

Block-sparse attention still spends quadratic time choosing which blocks to keep. PISA instead picks blocks by a pyramid Top- K search over $O(\log N)$ pooled key levels, using fused Triton kernels for training and inference. Against its baseline it is comparable on commonsense reasoning and better on retrieval. The abstract gives no speed numbers and no code link, so the wall-clock gain is unverified.

Why it matters. Block selection is the remaining quadratic step in long-context sparse attention; this removes it, if the kernels hold up.

3. Benchmarks, evaluations and reproductions

Benchmark	Sonnet 5.5	Opus 5.5	GPT-6 Sol	Reported by
AA Intelligence Index v4.3.2	56	58	48	Artificial Analysis
FrontierCode 1.1 (Main)	46.2%	54.4%	49.3%	Anthropic
GDPval-AA v2.1 (Elo)	1844	1846	1487	Anthropic
AA-Briefcase v1.1 (Elo)	1811	1822	1483	Anthropic
Chartography, no tools	61.6%	64.4%	53.6%	Anthropic
Terminal-Bench 4.0	70.6%	66.4%	—	Anthropic
OSWorld 2.1	80.1%	81.8%	—	Anthropic

Sources: row 1 [S2] [S6] [S4]; rows 2–7 from Anthropic’s Sonnet 5.5 table [S1]. Only row 1 is an independent measurement.

Only the first row is independent. On it, Opus 5.5 and Sonnet 5.5 sit 10 and 8 points above GPT-6 Sol (max). The remaining rows are Anthropic's own and include its choice of harness for the OpenAI column. Two harness gaps showed up this week. First, Chartography for Opus 5.5 is 64.4% "no tools" on the Sonnet page and 89.0% "with tools" on the Opus page [S5]. The benchmark and model are the same; only tool access differs. Second, Claude Opus 5 on AutomationBench v1.0.6 is 26.9% in both OpenAI's and Anthropic's tables [S3] [S5], but 50.3 in Xiaomi's [S8]. Cross-lab comparisons on agent benchmarks are not reliable until labs publish their harness settings.

4. Industry, funding and policy

- **UN Security Council, 10228th meeting (23 September).** The meeting was chaired by France, with Bengio, Altman, Amodei and Hugging Face's Delangue as briefers. Amodei proposed "a ban on using AI to make biological weapons", verification systems, and common testing standards with "a notification system for AI incidents". The White House's Michael Kratsios answered that international dialogue "cannot be allowed to drift towards global governance". [S15]
- **Ban Artificial Superintelligence Act (introduced 23 September).** Sen. Sanders and Rep. Casar's bill would ban AI that "exceeds human cognitive performance and capabilities across most domains", pause advanced development until a new cabinet-level Department of AI sets rules, and allow up to 20 years' imprisonment. It is a bill introduced by two members, not law. [S16]
- **Oregon Executive Order 26-26 (23 September).** The order directs the state CIO to draft third-party AI-safety review standards and to assess a "kill switch" requirement before agencies procure frontier models. The CIO's proposal is due within 90 days. It binds state procurement only. [S17]

5. What to watch next week

- **OpenAI DevDay, San Francisco, 29 September.** The event falls on this issue's publication date [S18]. What actually ships (as opposed to the rumoured lists circulating beforehand) is next week's lead.
- **Independent scores beyond Artificial Analysis for Sonnet 5.5 and GPT-6 Sol.** The vendor tables disagree on harness settings. The first third-party agent-benchmark runs will show whether Sonnet 5.5's Terminal-Bench lead over Opus 5.5 holds.
- **Wet-lab follow-up on ART.** The open question is whether any outside lab reports expression or activity data for the array-associated RTs, which is what would turn an agent-found anomaly into a finding.

6. Sources

#	Reference
S1	Anthropic. <i>Introducing Claude Sonnet 5.5</i> . 28 Sep 2026. https://www.anthropic.com/claude-sonnet-5-5
S2	Artificial Analysis. <i>Claude Sonnet 5.5</i> . Live page, accessed 29 Sep 2026. https://artificialanalysis.ai/models/claude-sonnet-5-5
S3	OpenAI. <i>Introducing GPT-6 Sol and Luna</i> . 22 Sep 2026. https://openai.com/index/introducing-gpt-6-sol-and-luna/
S4	Artificial Analysis. <i>GPT-6 Sol (max)</i> . Live page, accessed 29 Sep 2026. https://artificialanalysis.ai/models/gpt-6-sol
S5	Anthropic. <i>Introducing Claude Opus 5.5</i> . 22 Sep 2026. https://www.anthropic.com/claude-opus-5-5
S6	Artificial Analysis. <i>Claude Opus 5.5</i> . Live page, accessed 29 Sep 2026. https://artificialanalysis.ai/models/claude-opus-5-5
S7	Xiaomi MiMo. <i>MiMo-V2.6-Pro-MOPD model card</i> . 27 Sep 2026. https://huggingface.co/XiaomiMiMo/MiMo-V2.6-Pro-MOPD
S8	Xiaomi MiMo. <i>MiMo-V2.6-Pro-RL model card</i> . 21 Sep 2026. https://huggingface.co/XiaomiMiMo/MiMo-V2.6-Pro-RL
S9	Anthropic. <i>Claude discovers a novel enzyme system with CRISPR-like repeats</i> . 23 Sep 2026. https://www.anthropic.com/news/claude-discovers-novel-enzyme-system
S10	Yoon, P. H. et al. <i>Autonomous AI agents discover reverse transcriptases with tandem repeat arrays</i> . alphaXiv, 23 Sep 2026. https://www.alphaxiv.org/abs/2609.ai-agents-discover-reverse-transcriptases
S11	Zhou, Y. et al. <i>Just-in-Time Memory</i> . arXiv:2609.27334, 23 Sep 2026. https://arxiv.org/abs/2609.27334
S12	Patel, N. et al. <i>Learning to Discover Interesting Mathematics</i> . arXiv:2609.28603, 23 Sep 2026. https://arxiv.org/abs/2609.28603
S13	Tikhonov, P. et al. <i>Your Transformer Can Hold Two Thoughts at Once</i> . arXiv:2609.29845, 24 Sep 2026. https://arxiv.org/abs/2609.29845
S14	Tang, B. et al. <i>Block Sparse Attention with Log-Linear Complexity</i> . arXiv:2609.31093, 25 Sep 2026. https://arxiv.org/abs/2609.31093
S15	United Nations. <i>Security Council, 10228th meeting: AI and international security</i> . 23 Sep 2026. https://transcripts.un.org/en/sc/10228
S16	Office of Rep. Greg Casar. <i>Casar, Sanders introduce legislation to create new federal agency, ban artificial superintelligence</i> . 23 Sep 2026. https://casar.house.gov/media/press-releases/news-casar-sanders-introduce-legislation-create-new-federal-agency-ban
S17	Office of the Governor of Oregon. <i>Governor Kotek issues executive order to advance AI safety and oversight</i> . 23 Sep 2026. https://apps.oregon.gov/oregon-newsroom/OR/GOV/Posts/Post/governor-kotek-issues-executive-order-to-advance-ai-safety-and-oversight
S18	OpenAI. <i>Announcing OpenAI DevDay 2026</i> . https://openai.com/index/devday-2026/
S19	TechCrunch. <i>OpenAI launches GPT-6 Sol and Luna</i> . 22 Sep 2026. https://techcrunch.com/2026/09/22/openai-launches-gpt-6-sol-and-luna/

Compiled 29 September 2026. Every figure in this brief was read from the linked primary source on that date.
Corrections: Ali Habibullah.